Generative AI and LLM

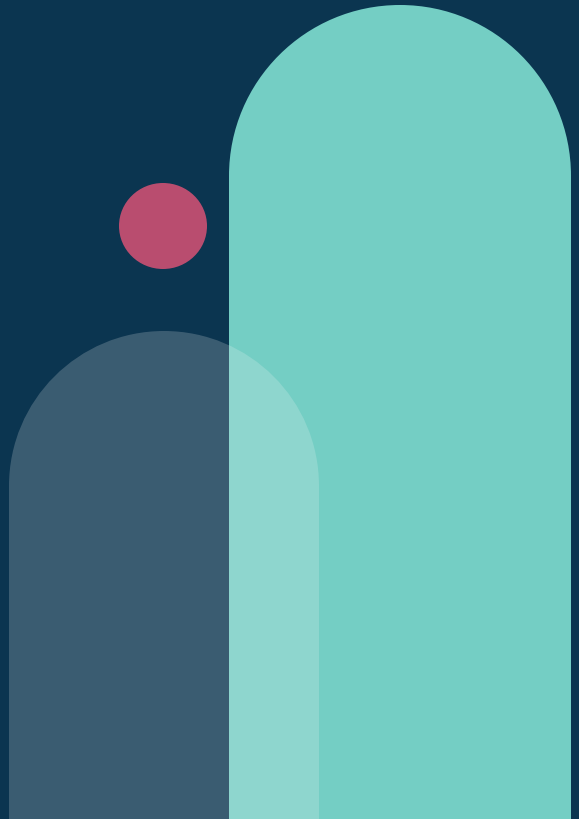
More on LLMs

Assistants

Popular LLM Providers

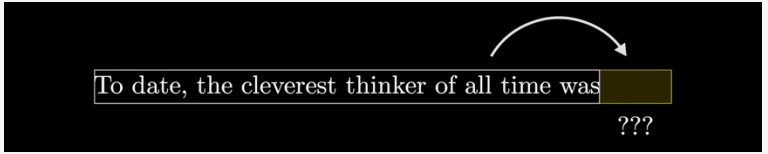
01

More on LLMs



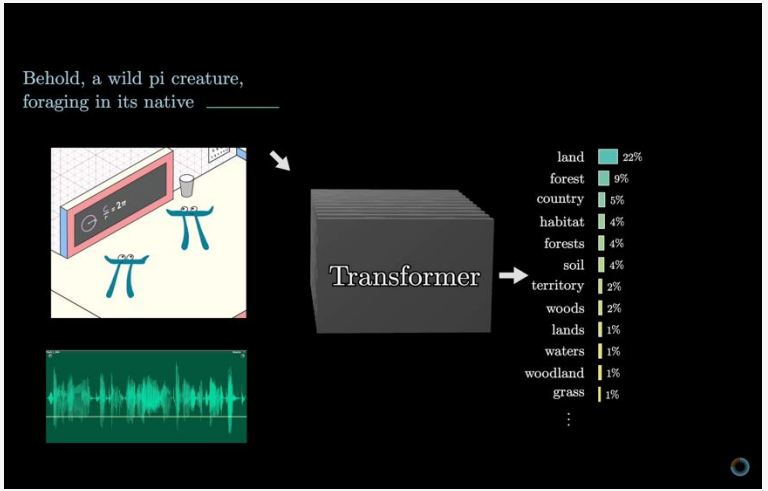
LLMs

Predict next word in a sequence



LLMs

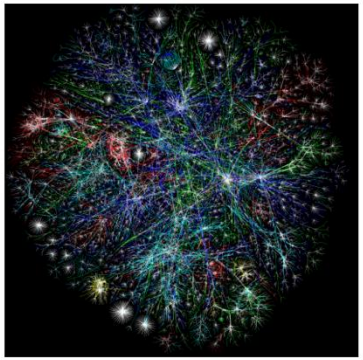
Generates text by predicting the **next word in a sequence**, informed by its training data. It typically based on transformer models, which excel at processing sequential data.



Where does the Knowledge come from?

Training them is more involved.

Think of it like compressing the internet.



Chunk of the internet,
~10TB of text



6,000 GPUs for 12 days, ~$2M
~1e24 FLOPS

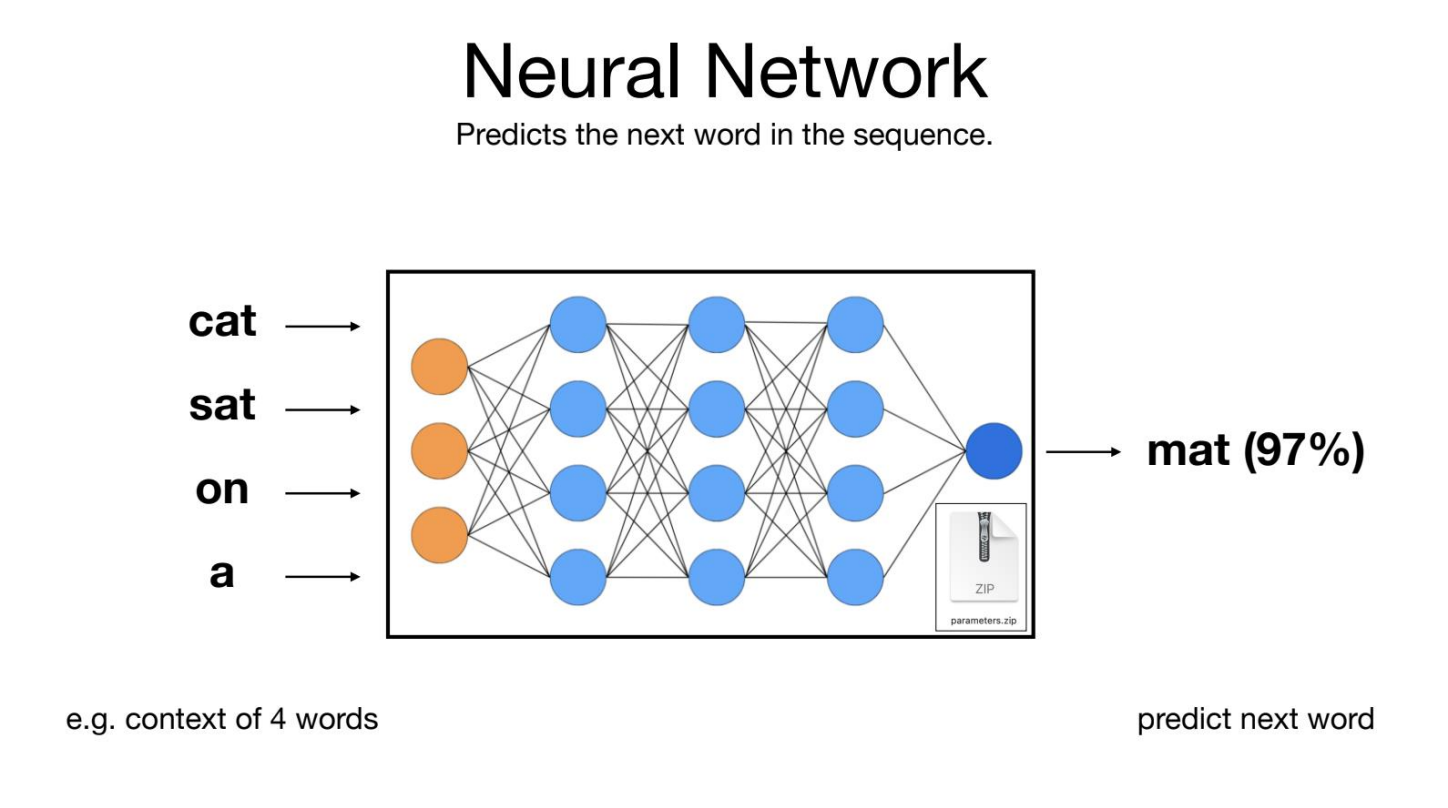


parameters.zip

~140GB file

*numbers for Llama 2 70B

How does it work?



Problem with Next Word Prediction?

Base LLM

Predicts next word, based on text training data

Once upon a time, there was a unicorn
that lived in a magical forest with all her unicorn friends



Problem with Base Model

Base LLM

Predicts next word, based on text training data

Once upon a time, there was a unicorn
that lived in a magical forest with all her unicorn friends

What is the capital of France?
What is France's largest city?
What is France's population?
What is the currency of France?





A base model doesn't interpret "instructions";
it simply generates related text based on its training data.

02

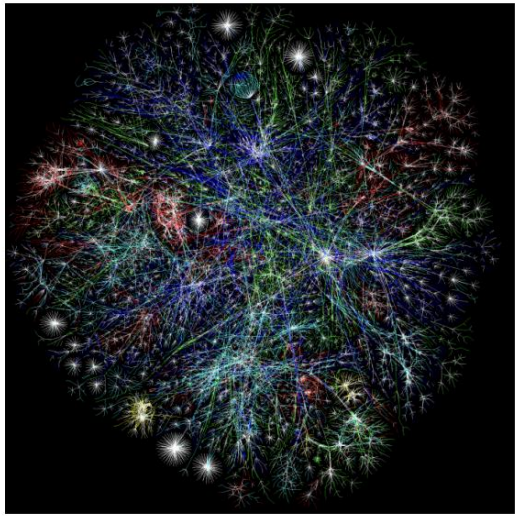
Assistants

Instruct Model (Assistant)

Training the Assistant

(Just swap the dataset, then continue training)

written by people:



<USER>

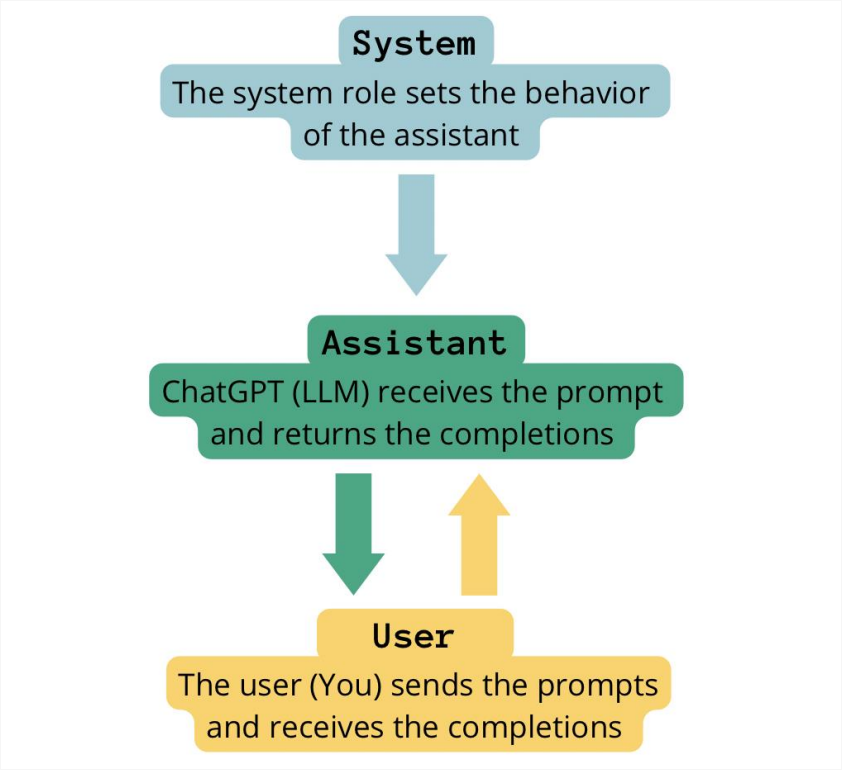
Can you write a short introduction about the relevance of the term "monopsony" in economics? Please use examples related to potential monopsonies in the labour market and cite relevant research.

<ASSISTANT>

"Monopsony" refers to a market structure where there is only one buyer for a particular good or service. In economics, this term is particularly relevant in the labor market, where a monopsony employer has significant power over the wages and working conditions of their employees. The presence of a monopsony can result in lower wages and reduced employment opportunities for workers, as the employer has little incentive to increase wages or provide better working conditions…

~100K conversations

Roles in LLM




```
messages =
[
    system
    user
    assistant
    user
    assistant
    ...
]
```

Instruct Model (Assistant)

After finetuning you have an Assistant:

<USER>

Can you help me with this code? It seems like there is a bug.
print(“hello world)

<ASSISTANT>

It looks like you forgot to close the string passed to the function print. You have to add a closing quote to properly terminate the string. Here is the corrected function:

print(“hello world”)

Let me know if I can help with anything else!

Base Model vs Instruct Model

Two Types of large language models (LLMs)

Base LLM

Predicts next word, based on text training data

Once upon a time, there was a unicorn
that lived in a magical forest with
all her unicorn friends

What is the capital of France?
What is France's largest city?
What is France's population?
What is the currency of France?

Instruction Tuned LLM

Tries to follow instructions

Fine-tune on instructions and
good attempts at following those instructions.

RLHF: Reinforcement Learning with
Human Feedback

Helpful, Honest, Harmless

What is the capital of France?
The capital of France is Paris.

Train an Assistant

Summary: how to train your ChatGPT



every ~year

Stage 1: Pretraining
1. Download ~10TB of text.
2. Get a cluster of ~6,000 GPUs.
3. Compress the text into a neural network, pay ~$2M, wait ~12 days.
4. Obtain **base model**.



every ~week

Stage 2: Finetuning
1. Write labeling instructions
2. Hire people (or use scale.ai!), collect 100K high quality ideal Q&A responses, and/or comparisons.
3. Finetune base model on this data, wait ~1 day.
4. Obtain **assistant model.**
5. Run a lot of evaluations.
6. Deploy.
7. Monitor, collect misbehaviors, go to step 1.

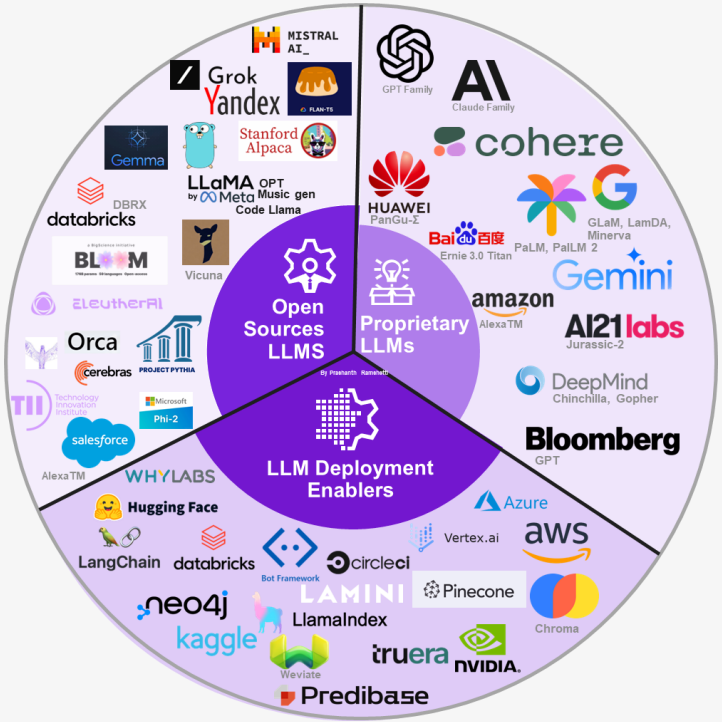
How to access LLMs?

Open Source LLMs

- **Direct Download**: Access models by downloading pre-trained weights and running them locally on your own hardware. Ollama, LMStudio etc.
- **ML Frameworks**: Utilize frameworks like TensorFlow or PyTorch to fine-tune and deploy models, whether open-source or proprietary, on your infrastructure.
- **Containerization and Orchestration**: Deploy LLMs using Docker and Kubernetes, which enable scalable and efficient management of models in production environments.
- **Community Platforms**: Leverage platforms like Hugging Face's Model Hub, where you can easily access, fine-tune, and deploy open-source LLMs.

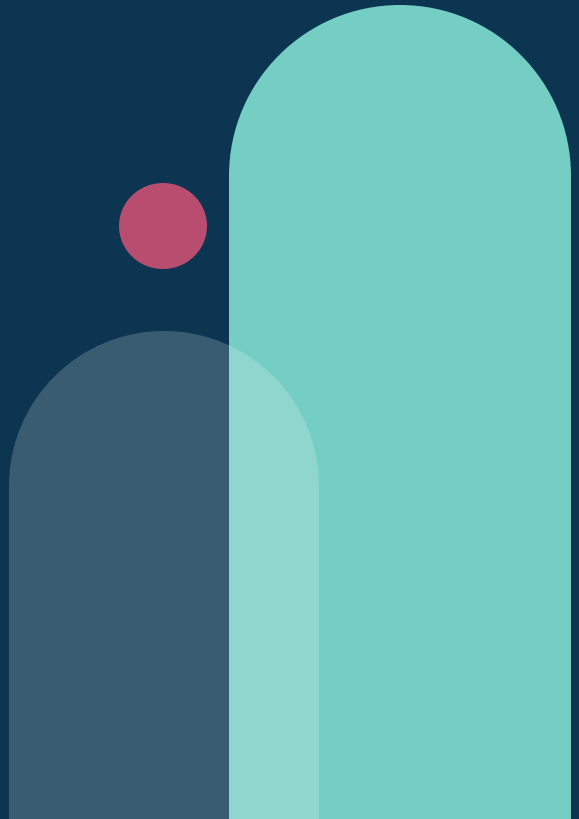
Proprietary LLMs

- **API Access**: Use APIs provided by companies like OpenAI or Google to interact with LLMs such as GPT-4 or Gemini, typically requiring a subscription or pay-per-use model.
- **Integrated Cloud Services**: Access proprietary LLMs via cloud platforms like Azure or AWS, which offer integrated services and deployment options for enterprise use.

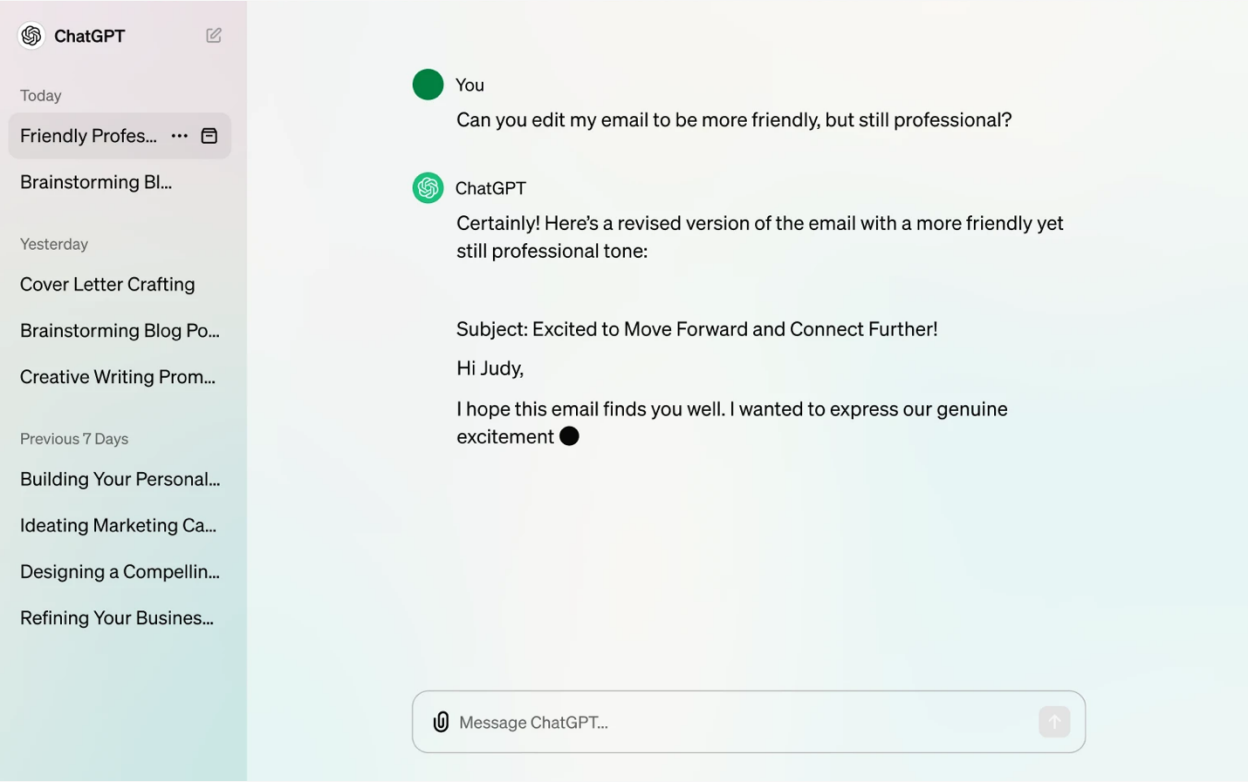


03

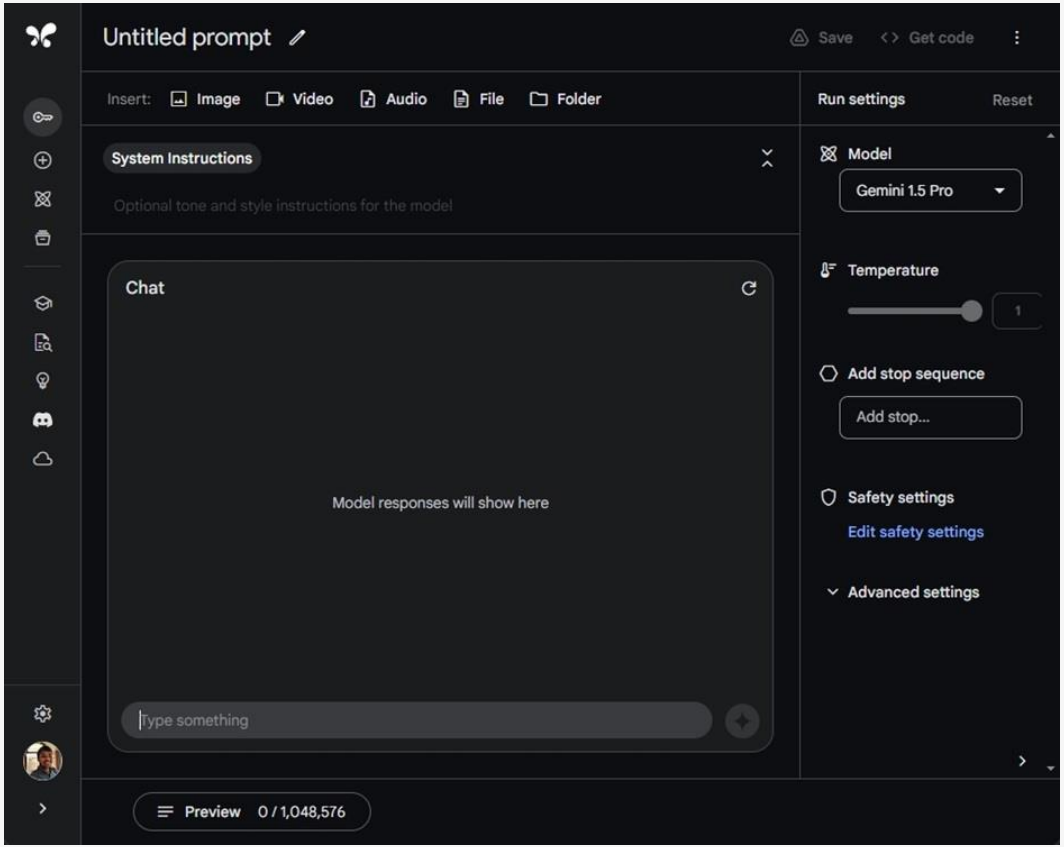
Popular LLM Providers



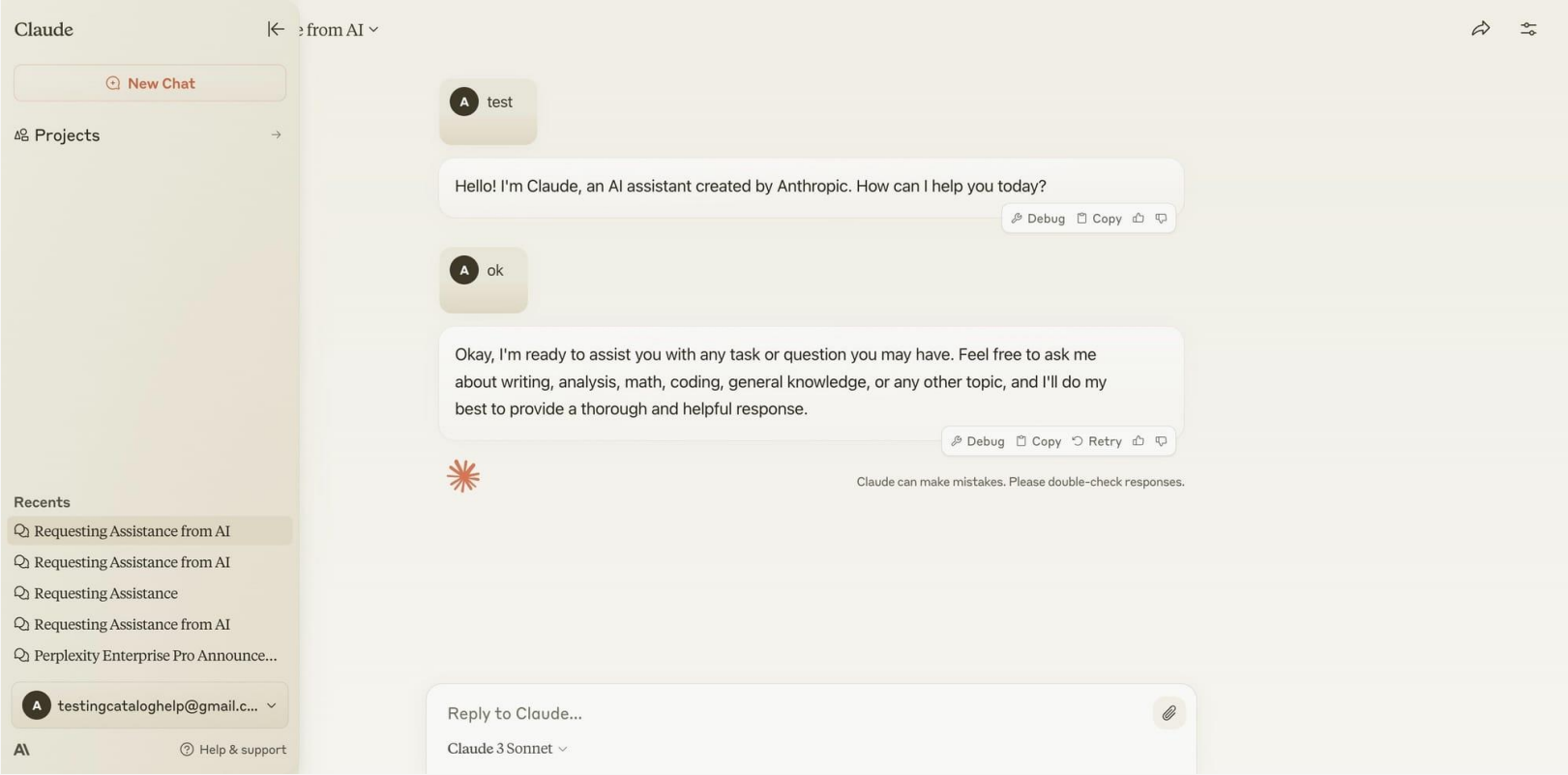
ChatGPT


ChatGPT
Today
Friendly Profes...
Brainstorming Bl...
Yesterday
Cover Letter Crafting
Brainstorming Blog Po...
Creative Writing Prom...
Previous 7 Days
Building Your Personal...
Ideating Marketing Ca...
Designing a Compellin...
Refining Your Busines...
You
Can you edit my email to be more friendly, but still professional?
ChatGPT
Certainly! Here's a revised version of the email with a more friendly yet still professional tone:
Subject: Excited to Move Forward and Connect Further!
Hi Judy,
I hope this email finds you well. I wanted to express our genuine excitement
Message ChatGPT...

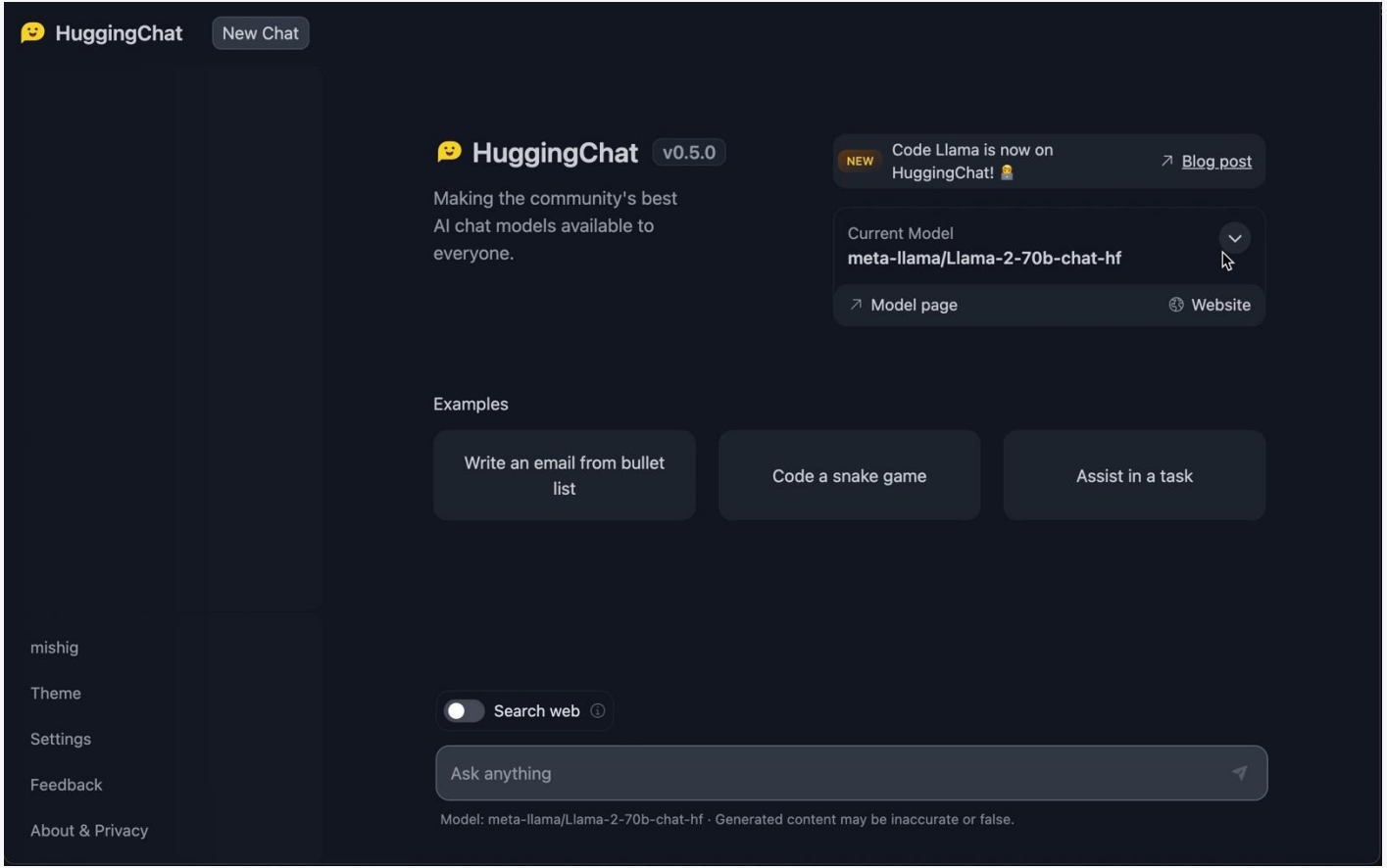
Google AI Studio



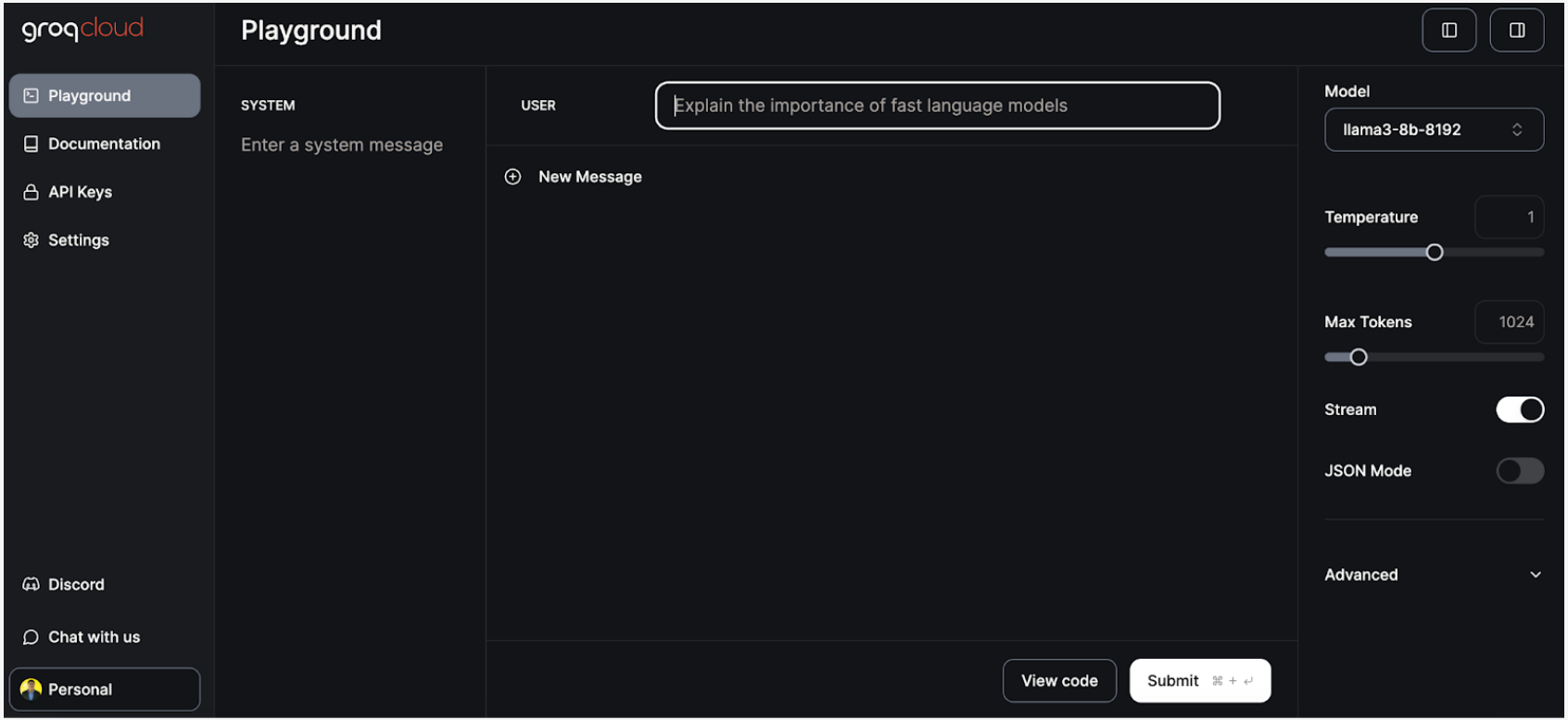
Claude


Claude
from AI
New Chat
Projects
test
Hello! I'm Claude, an AI assistant created by Anthropic. How can I help you today?
Debug
Copy
ok
Okay, I'm ready to assist you with any task or question you may have. Feel free to ask me about writing, analysis, math, coding, general knowledge, or any other topic, and I'll do my best to provide a thorough and helpful response.
Debug
Copy
Retry
Claude can make mistakes. Please double-check responses.
Recents
Requesting Assistance from AI
Requesting Assistance from AI
Requesting Assistance
Requesting Assistance from AI
Perplexity Enterprise Pro Announce...
testingcataloghelp@gmail.c...
Help & support
Reply to Claude...
Claude 3 Sonnet

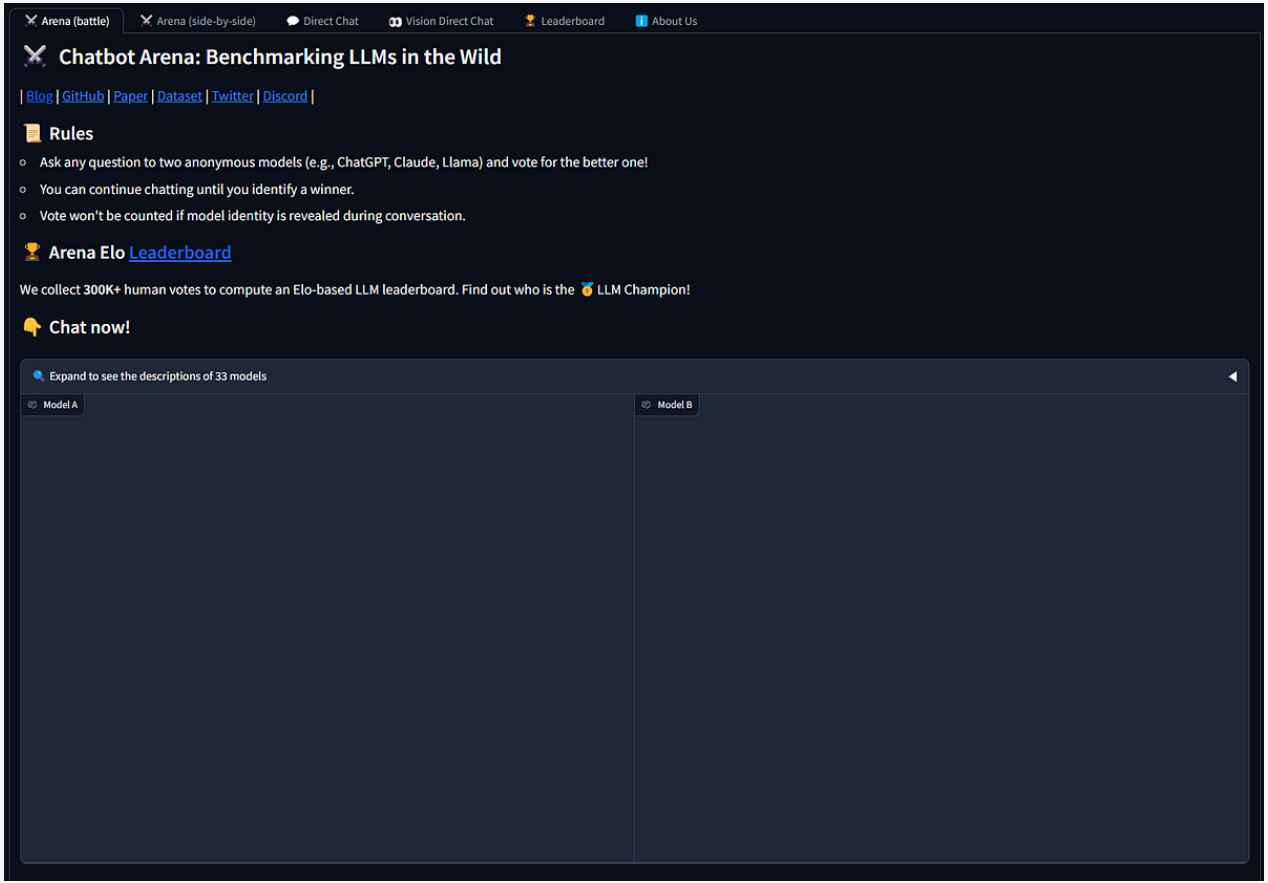
HuggingChat



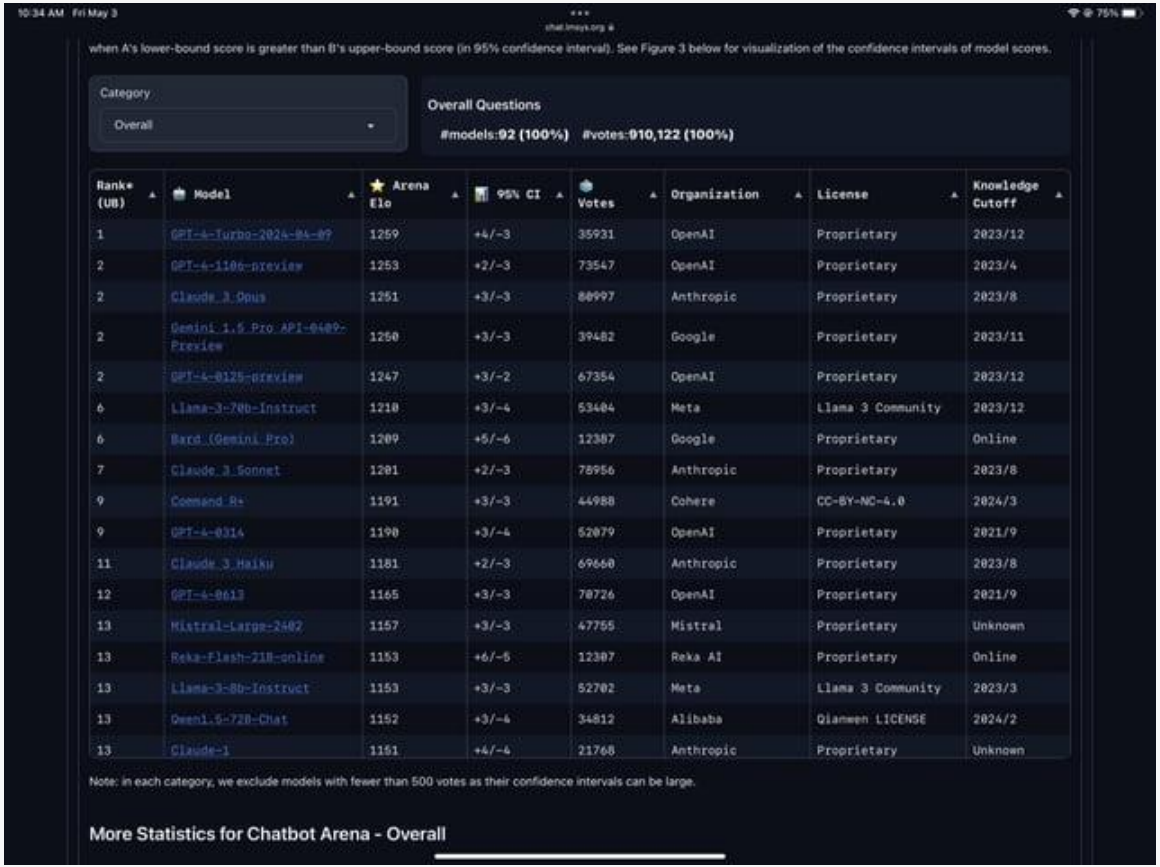
Groq


groqcloud
Playground
Playground
Documentation
API Keys
Settings
Discord
Chat with us
Personal
SYSTEM
Enter a system message
USER
Explain the importance of fast language models
New Message
View code
Submit
Model
llama3-8b-8192
Temperature
1
Max Tokens
1024
Stream
JSON Mode
Advanced

LMSys


Arena (battle)
Arena (side-by-side)
Direct Chat
Vision Direct Chat
Leaderboard
About Us
Chatbot Arena: Benchmarking LLMs in the Wild
| Blog | GitHub | Paper | Dataset | Twitter | Discord |
Rules
Ask any question to two anonymous models (e.g., ChatGPT, Claude, Llama) and vote for the better one!
You can continue chatting until you identify a winner.
Vote won't be counted if model identity is revealed during conversation.
Arena Elo Leaderboard
We collect 300K+ human votes to compute an Elo-based LLM leaderboard. Find out who is the LLM Champion!
Chat now!
Expand to see the descriptions of 33 models
Model A
Model B

LMSys Leaderboard



when A's lower-bound score is greater than B's upper-bound score (in 95% confidence interval). See Figure 3 below for visualization of the confidence intervals of model scores.

Category: Overall

Overall Questions

#models:92 (100%) #votes:910,122 (100%)

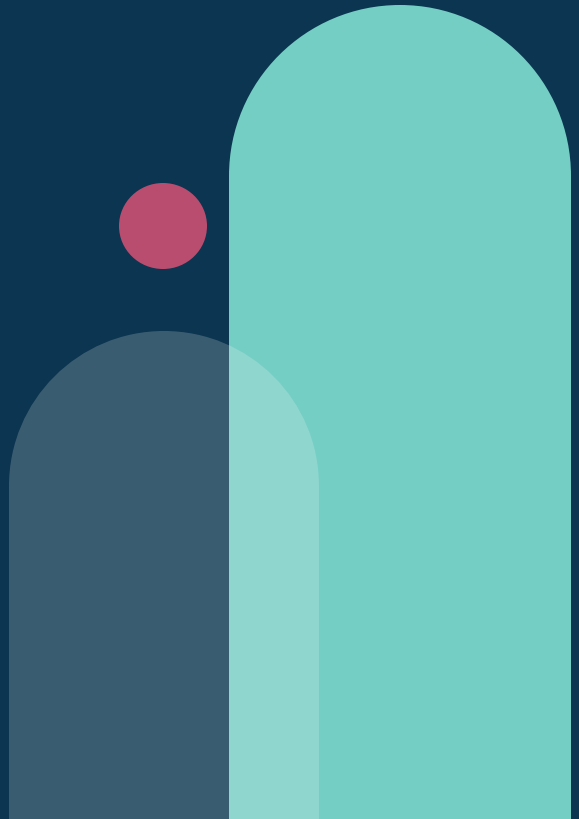
Rank* (UB)	Model	Arena Elo	95% CI	Votes	Organization	License	Knowledge Cutoff
1	GPT-4-Turbo-2024-04-09	1259	+4/-3	35931	OpenAI	Proprietary	2023/12
2	GPT-4-1106-preview	1253	+2/-3	73547	OpenAI	Proprietary	2023/4
2	Claude 3 Opus	1251	+3/-3	80997	Anthropic	Proprietary	2023/8
2	Gemini 1.5 Pro API-0409-Preview	1250	+3/-3	39482	Google	Proprietary	2023/11
2	GPT-4-0125-preview	1247	+3/-2	67354	OpenAI	Proprietary	2023/12
6	Llama-3-70b-Instruct	1210	+3/-4	53404	Meta	Llama 3 Community	2023/12
6	Bard (Gemini Pro)	1209	+5/-6	12387	Google	Proprietary	Online
7	Claude 3 Sonnet	1201	+2/-3	78956	Anthropic	Proprietary	2023/8
9	Command R+	1191	+3/-3	44988	Cohere	CC-BY-NC-4.0	2024/3
9	GPT-4-0314	1190	+3/-4	52079	OpenAI	Proprietary	2021/9
11	Claude 3 Haiku	1181	+2/-3	69660	Anthropic	Proprietary	2023/8
12	GPT-4-0613	1165	+3/-3	70726	OpenAI	Proprietary	2021/9
13	Mistral-Large-2402	1157	+3/-3	47755	Mistral	Proprietary	Unknown
13	Reka-Flash-21B-online	1153	+6/-6	12387	Reka AI	Proprietary	Online
13	Llama-3-8b-Instruct	1153	+3/-3	52702	Meta	Llama 3 Community	2023/3
13	Qwen1.5-72B-Chat	1152	+3/-4	34812	Alibaba	Qianmen LICENSE	2024/2
13	Claude-1	1151	+4/-4	21768	Anthropic	Proprietary	Unknown

Note: in each category, we exclude models with fewer than 500 votes as their confidence intervals can be large.

More Statistics for Chatbot Arena - Overall

04

Access through Code



Thanks!

Do you have any questions?

